

Equation Sheet

Circular Motion:

$$F_{net} = ma \quad a_c = \frac{v^2}{r} \quad F_{net,c} = ma_c = m \left(\frac{v^2}{r} \right) \quad 1 \text{ rev} = 2\pi r = \pi D \quad g's = \frac{F_a}{F_g}$$

Universal Gravitation:

$$F_g = \frac{Gm_1m_2}{d^2} \quad a_g = \frac{Gm_0}{d^2} \quad v_{orbit} = \sqrt{\frac{Gm_0}{d}} \quad T_{orbit} = \frac{2\pi d}{v_{orbit}}$$

Electricity & Circuits

$$1 e^- = -1.6 \times 10^{-19} C \quad 6.25 \times 10^{18} e^- = 1 C \quad \text{Mass } e^- = 9.11 \times 10^{-31} kg \quad K_c = 9 \times 10^9 \frac{N \cdot m^2}{C^2}$$

$$E = \frac{K_c q_s}{d^2} \quad F_E = \frac{K_c q_1 q_2}{d^2} = qE$$

$$F_E = qE \quad V = \frac{\text{Energy}}{q} \quad I = \frac{q}{t} \quad V = IR \quad P = \frac{\text{Energy}}{t} \quad P = IV \quad P = I^2 R \quad P = \frac{V^2}{R}$$

$$R_{series} = R_1 + R_2 + R_3 + \dots + R_n \quad R_{parallel} = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots + \frac{1}{R_n}}$$

Work/Energy/Power

$$\text{Work} = \Delta \text{Energy} \quad \text{Work} = F \cdot d \cdot \cos\theta \quad \text{Power} = \frac{\text{Work}}{\text{Time}} = \frac{\Delta \text{Energy}}{\text{Time}}$$

$$F_s = kx \quad GPE = mgh \quad KE = \frac{1}{2}mv^2 \quad EPE = \frac{1}{2}kx^2$$

Waves

$$T = \frac{1}{f} \quad f = \frac{1}{T} \quad v_{wave} = \frac{\lambda}{T} = \lambda f$$

$$f_{heard} = f_{source} \left\{ \frac{v_{sound}}{v_{sound} - v_{source}} \right\}$$

*for a moving source

Motor Project

$$P_{out} = \frac{\text{Work}}{t} = \frac{\Delta GPE}{t} = \frac{mgh}{t} \quad P_{in} = IV$$

$$\eta = \frac{P_{out}}{P_{in}} \quad \tau = F_g r = mgr \quad (g = 9.8 \frac{m}{s^2})$$

Constants/Conversions:

$$a_g \text{ or } g = 9.81 \text{ m/s}^2$$

$$G = 6.673 \times 10^{-11} \frac{Nm^2}{kg^2}$$

$$K_c = 9 \times 10^9 N \cdot m^2 / C^2$$

$$M_{earth} = 5.98 \times 10^{24} \text{ kg}$$

$$r_{earth} = 6.38 \times 10^6 \text{ m}$$

$$\text{Mass } e^- = 9.11 \times 10^{-31} \text{ kg}$$

$$1 e^- = -1.6 \times 10^{-19} C \quad 6.25 \times 10^{18} e^- = 1 C$$

$$1 \text{ revolution} = 2\pi r$$

$$1 \text{ hp} = 746 \text{ W}$$

Metric Prefixes

k	kilo	10 ³
	base	10 ⁰
c	centi	10 ⁻²
m	milli	10 ⁻³
μ	micro	10 ⁻⁶
n	nano	10 ⁻⁹
p	pico	10 ⁻¹²